

Tony Hoare: British Computer Scientist. ∴ Turing Award.

① Quick Sort: Elegant Algorithm.

② Formal verification. $\boxed{\{P\} S \{Q\}}$

Hoare's triplet:

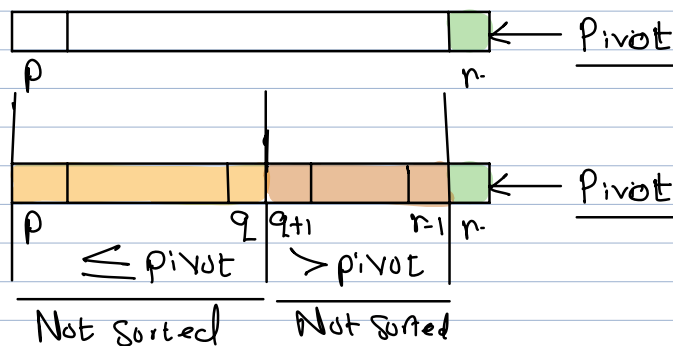
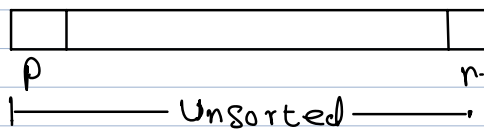
③ C.S.P. = Communicating Sequential Processes.

Concurrent Programming.

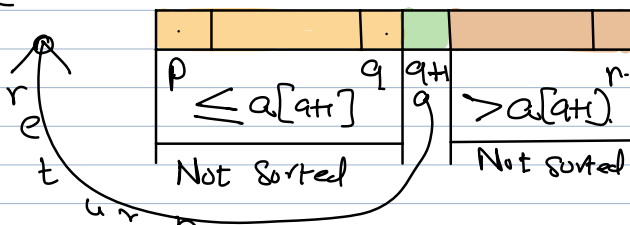
Go: C.S.P. 25/30

channel Go-routine
Non-shared-state-concurrency.

PARTITION (a, p, r) $0 \leq p < r < N$.



$a(q)$



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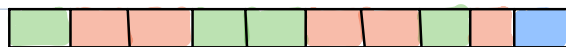
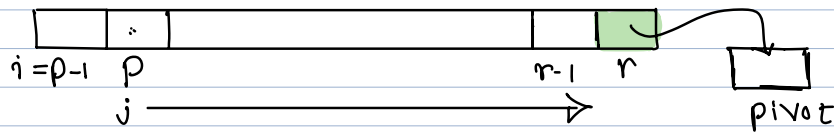
void quick_sort (int* a, int p, int r)
{
    if (p < r)
    {
        int q = partition (a, p, r);
        quick_sort (a, p, q-1);
        quick_sort (a, q+1, r);
    }
}

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int partition (int* a, int p, int r)
{
    int pivot = a[r];
    int j; // j = p to r-1
}

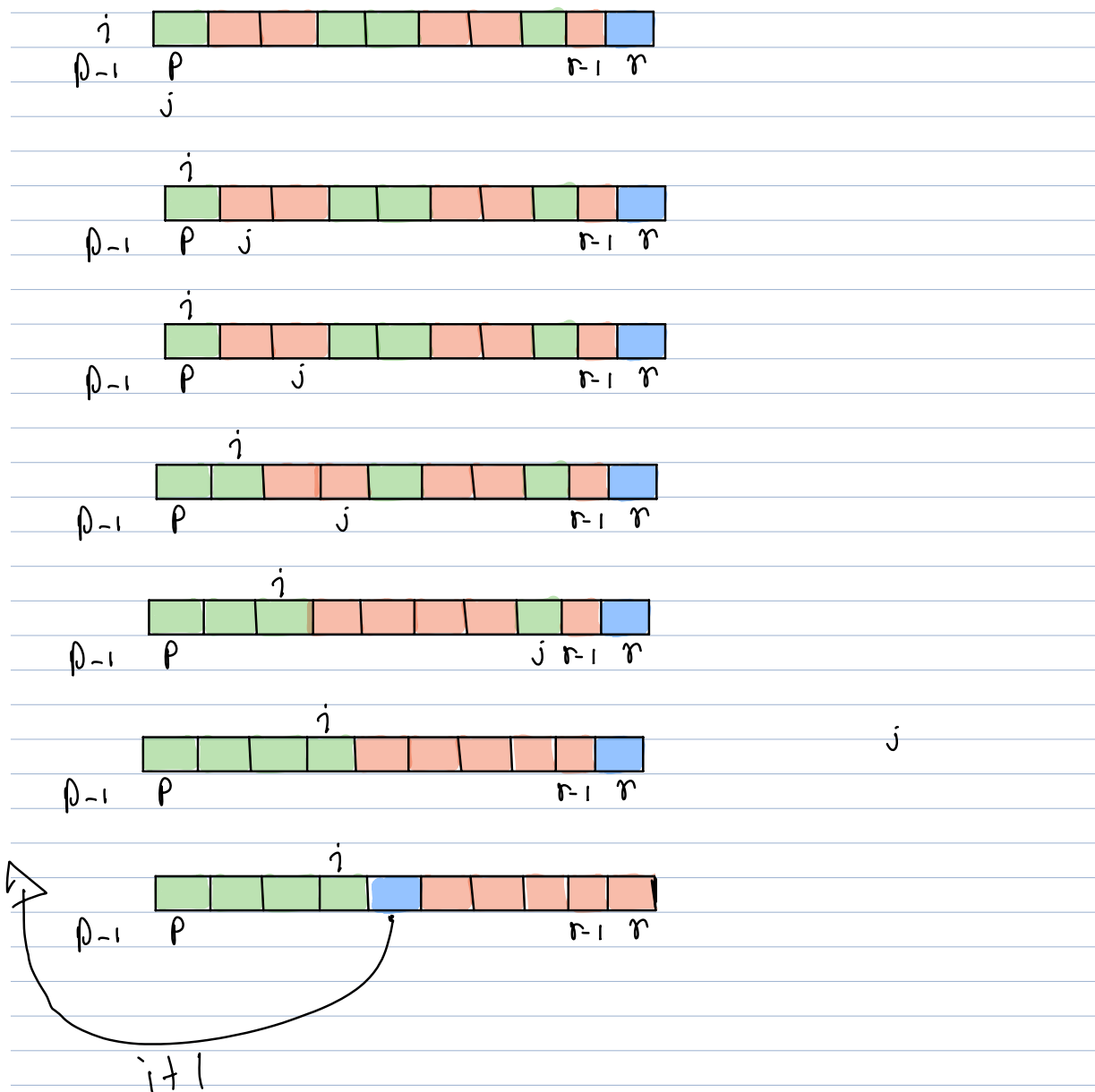
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for j = p to j = r-1
    if a[j] <= pivot
        i = i + 1
        a[i] ↔ a[j]

```



Loop-Invariant of Partition Function.

$$\forall j [p \leq j < r] \quad \forall k [p \leq k \leq i] \quad a[k] \leq \text{pivot}.$$

$$\forall j [p \leq j < r] \quad \forall k [i+1 \leq k < j] \quad a[k] > \text{pivot}.$$

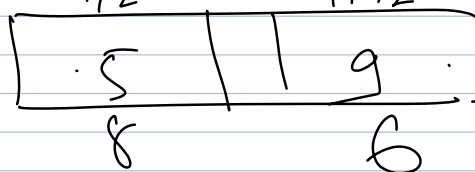
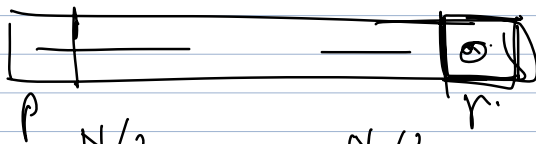
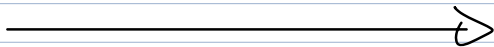
$$\forall j \quad p \leq j < r \rightarrow \exists i \quad \left(\begin{array}{l} p \leq k \leq i \rightarrow a[k] \leq \text{pivot} \\ \wedge \\ i+1 \leq k < j \rightarrow a[k] > \text{pivot} \end{array} \right)$$

$$i = p-1 \quad j = p$$

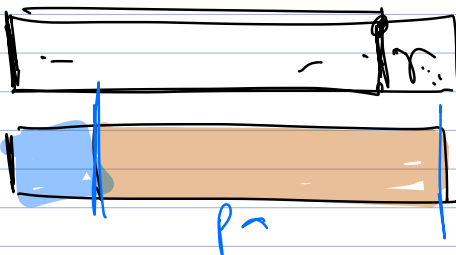
$$\begin{array}{l} p \text{ to } i \leq \text{pivot} \rightarrow p \text{ to } p-1 \leq \text{pivot} \\ i+1 \text{ to } j-1 > \text{pivot} \rightarrow p \text{ to } p-1 > \text{pivot} \end{array}$$

$$\begin{array}{ccc} j_0 & i_0 & a[r_0] \\ p \leq \text{pivot} & i_0 & j_0-1 \\ & & \boxed{} \\ & & j_0 \end{array}$$

Adversary Input.



$$n^2$$



$$n \lg n$$

$$n^2$$